

Model Order Reduction techniques

Assignments - Part 2

Nicola R. Franco, Andrea Manzoni

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Students are required to pick (and elaborate upon) any of the three problems reported in the following pages.

All projects have a common skeleton. In particular, all case studies come with a suitable dataset containing a list of pre-computed FOM simulations. A showoff of the three case studies is available at the following link

https://colab.research.google.com/drive/1_kAQRJD0iv4ER2UosuTAClIPB0_yJUW?usp=sharing

The same notebook can be used to access and inspect the data.

Starting from there, students are requested to

- Implement, train and assess a data-driven ROM of choice (POD-NN, DL-ROM or, when appropriate, AE+SINDy);
- Exploit the ROM to answer a given question.

Students are expected to present their work by synthesizing the main steps in a suitable IPython notebook. To foster readability, technical pieces of code should be placed either outside of the notebook (i.e. within external ".py" files) or in suitable code sections that can be hidden/toggled by hand. To ensure a smooth online execution of the notebook, trained ROMs can be saved (once and for all) and later loaded on the fly.

The IPython notebook should report/contain:

- i) a brief discussion about the design rationale, namely: why are we resorting to data-driven ROMs here? What is stopping us from using plain POD-Galerkin? Why did you pick a certain approach over the others¹? How did you choose the latent/reduced dimension?;
- ii) a ROM solver that can be queried in real-time, together with a short summary about its performances (computational cost, accuracy);
- iii) any additional results concerning project-specific tasks.

¹Or, in case you've tested multiple approaches and then simply selected the best one: can you explain why a certain ROM ended up working better than the others?

Material design for a stunt training facility

(linear elasticity, time-dependent)

A company is developing a high-performance flooring system to enhance safety in stunt training facilities. The design features a dual-layer structure: the bottom layer is stiffer, providing essential stability, while the top layer is softer, offering improved shock absorption to minimize the impact of falls.

As a part of the design process, the engineers working at the company are looking for the best compromise in terms of stability and impact absorption, trying to come up with the correct proportions between the two layers. To this, numerical simulations are being carried out.

Focusing on an ideal, isolated, portion of the floor, the engineers are modeling the behavior of the floor using the equations of linear elasticity. For simplicity, the analysis is conducted in two dimensions. Specifically, let $\Omega = (0, 1)^2$ denote the portion of the floor under study. Let

$$\mathbf{f}(t) := \begin{cases} [0, -5 \sin(2\pi t)]^\top & 0 \leq t \leq 0.5 \\ [0, 0]^\top & t > 0.5 \end{cases}$$

be an external force representing the localized vertical impact exerted by an athlete's foot during contact with the floor. Set the final time to be $T = 1$. Let $\mathbf{u} : \Omega \times [0, T] \rightarrow \mathbb{R}^2$ be the vector-field describing the local displacement of the floor over time. We model $\mathbf{u}$ as the solution to

$$\begin{cases} \frac{\partial^2 \mathbf{u}}{\partial t^2} = \nabla \cdot [\lambda \nabla \cdot \mathbf{u} + \mu (\nabla \mathbf{u} + \nabla \mathbf{u}^\top)] + \mathbf{g} & \text{in } \Omega \times (0, T] \\ \mathbf{u}(\cdot, 0) \equiv \mathbf{0} & \text{in } \Omega \\ \mathbf{u} \equiv \mathbf{0} & \text{on } \Gamma_{\text{bottom}} \times [0, T] \\ \lambda (\nabla \cdot \mathbf{u}) \cdot \mathbf{n} + \mu (\nabla \mathbf{u} + \nabla \mathbf{u}^\top) \cdot \mathbf{n} \equiv \mathbf{f} & \text{on } \Gamma_{\text{top}} \times [0, T] \\ \lambda (\nabla \cdot \mathbf{u}) \cdot \mathbf{n} + \mu (\nabla \mathbf{u} + \nabla \mathbf{u}^\top) \cdot \mathbf{n} \equiv \mathbf{0} & \text{on } \partial\Omega \setminus (\Gamma_{\text{bottom}} \cup \Gamma_{\text{top}}) \times [0, T] \end{cases} \quad (4)$$

where $\mathbf{g} = [0, -1]^\top$ is a fixed body force (caused by the floor own weight). The Lamé coefficients of the material are given as

$$\lambda(x, y) = \begin{cases} 10 & y \geq \delta \\ 50 & y < \delta \end{cases} \quad \text{and} \quad \mu(x, y) = \begin{cases} 10 & y \geq \delta \\ 50 & y < \delta, \end{cases}$$

where $\delta \in [0.5, 0.9]$ is a parameter describing the thickness of the two layers composing the floor.

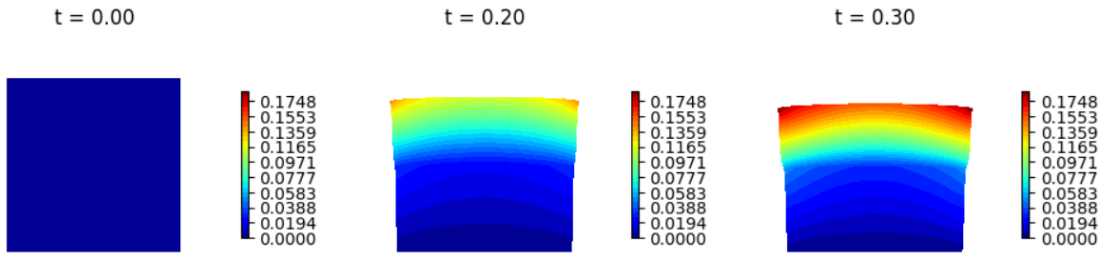


Figure 3. Three time steps of a FOM simulation, here obtained for $\delta = 0.5305$.

Tasks

Let $\boldsymbol{\mu} := \delta$ be the 1-dimensional parameter describing the system¹. You are given access² to a dataset consisting of $N = 100$ pre-computed random simulations, $\{\boldsymbol{\mu}_i, \mathbf{u}_{\boldsymbol{\mu}_i}\}_{i=1}^N$, obtained for different values of the problem parameter. The dataset was constructed by drawing random values of $\boldsymbol{\mu}$, independently and uniformly in the parameter space, and by solving the corresponding equations of linear elasticity using Finite Element in space and finite differences in time. In particular, each PDE solution is being represented as a matrix, $\mathbf{u}_{\boldsymbol{\mu}_i} \cong u_{\boldsymbol{\mu}_i}$, listing $N_t = 51$ time snapshots equally spaced in time ($\Delta t = 0.02$), each with $N_h = 3362$ degrees of freedom.

- i) Split the data in two blocks, keeping 75 simulations for training and 25 for testing. Choose a suitable approach between POD-NN, DL-ROM and AE+SINDy, and implement the corresponding reduced order model. Evaluate the performances of the ROM in terms of the following error metric (test average of relative space-time L^2 errors),

$$\frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} \left[\frac{\sum_{j=1}^{N_t} \|\mathbf{u}_{i,j}^{\text{test}} - \tilde{\mathbf{u}}_{i,j}^{\text{test}}\|_{L^2(\Omega)}}{\sum_{j=1}^{N_t} \|\mathbf{u}_{i,j}^{\text{test}}\|_{L^2(\Omega)}} \right], \quad (5)$$

where $\mathbf{u}_{i,j}^{\text{test}}$ is the j th time stamp of the i th test simulation and $\tilde{\mathbf{u}}_{i,j}^{\text{test}}$ is the corresponding ROM approximation. You may consider your ROM to be satisfactory if the error in Eq. (5) is **below 2.5%**.

- ii) Considering that the FOM requires ≈ 8.11 seconds to generate a single trajectory, quantify the computational speed-up provided by the ROM.

We are interested in studying how the thickness of the rigid layer, δ , affects the maximum deformation

$$J(\mathbf{u}) := \max_{(\mathbf{x}, t) \in \Omega \times [0, T]} \|\mathbf{u}(\mathbf{x}, t)\|.$$

- iii) Leveraging the ROM, draw a plot of the map $\delta \mapsto J(\mathbf{u}_\delta)$. To foster interpretability, include also the values of J obtained by directly evaluating the functional over the training data and the test data (e.g., using "black dots" for training samples and "red dots" for test instances).

To gain additional insights, the company also wants to study the behavior of the maximum deformation over time, given by the time-dependent functional

$$J_t(\mathbf{u}) := \max_{\mathbf{x} \in \Omega} \|\mathbf{u}(\mathbf{x}, t)\|.$$

- iv) Pick 1001 equally spaced values of δ in $[0.5, 0.9]$. For each value of δ , use the ROM to draw a plot³ of the map $t \mapsto J_t(\mathbf{u}_\delta)$. Does the time of maximum deformation change with δ ? How?

¹Even though we are in the scalar case, $p = 1$, we use the bold notation to avoid confusion with the shear modulus μ .

²See the [auxiliary notebook](#)

³We can place all curves within the same plot, e.g. by letting the color fade from red to blue as you change the value of δ .